9 You are given that

$f(x) = 2x + 3$ for $x < 0$ and

$g(x) = x^2 - 2x + 1$ for $x > 1$.

(a) Find $gf(x)$, stating the domain. **[3]**

(b) State the range of $gf(x)$. **[1]**

(c) Find $(gf)^{-1}(x)$. **[5]**